Part 1: Preprocessing

Filter-bank CSP + band power in short time windows

ML Frontiers, VISTEC · BCIC IV-2a, left vs right hand MI

Why not just use CSP, or MixNet's input?

- MI shows up as ERD/ERS in Mu and beta, around C3/C4 [lecture p.51]
- Classic CSP keeps one log-variance per trial, so it throws away the timing inside the trial [lecture p.62]
- MixNet keeps the whole CSP time series [lecture p.63]: 7,200 numbers per trial, learned from only ∼115 training trials

	Classic FBCSP	MixNet	Ours
Keeps timing	no	all 400 samples	15 windows
Shape per trial	36	18×400	36×15
Numbers per trial	36	7,200	540

Hypothesis

Band power in short time windows keeps the timing of ERD/ERS, at a size a small model can learn from ∼115 trials.

Pipeline: four steps, two of them learned



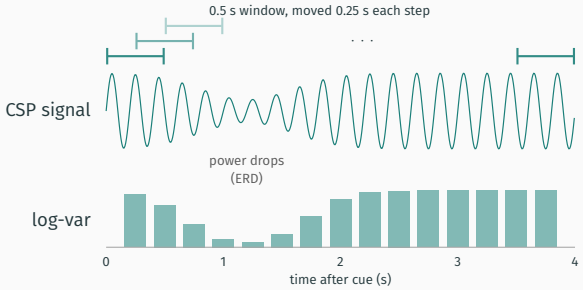
Teal boxes: fit on the training fold only

- "Fit CSP: training data only" [lecture p.60]
- "Training-set per-channel standardization" [lecture p.31]
- Validation and session E only reuse the saved filters and mean/SD. A self-check in the code tests this.

Output per trial

36 features (9 bands × 4 CSP filters) × 15 time windows, standardized

Our change: one log-variance per short window



Normalized log-variance

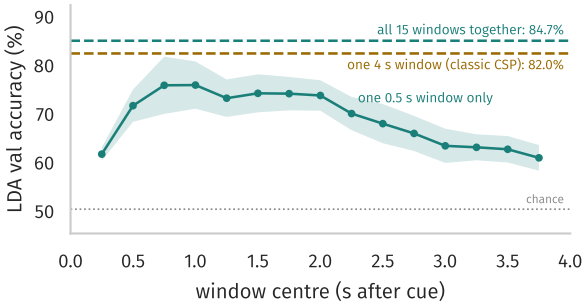
For band b, CSP filter i, window w:

$$f_{b,i,w} = \log \frac{\operatorname{var}(z_{b,i}[w])}{\sum_{j=1}^{4} \operatorname{var}(z_{b,j}[w])}$$

[lecture p.59]

- $(400 - 50)/25 + 1 = 15$ windows in 4 s
- This step learns nothing, so it cannot leak
- Then each feature is standardized with the training-fold mean and SD [lecture p.31]

Quick LDA test: class information is not spread evenly in time

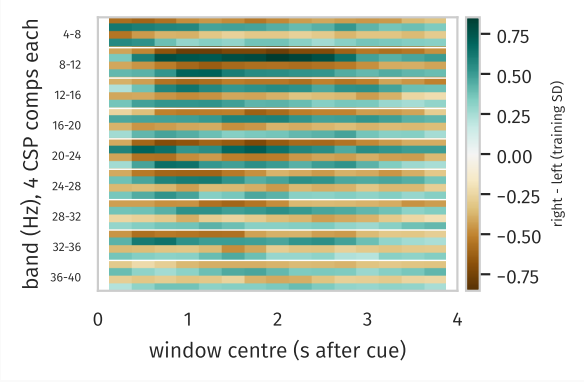


- One 0.5 s window (36 numbers, one LDA per window): peaks at 0.75-1 s (75.5%), near 61% at both ends
- One 4 s window = whole trial, classic CSP (36 numbers): 82.0%
- All 15 windows together = our features (540 numbers): 84.7%, better in 8 of 9 subjects

How we got these numbers

A simple shrinkage LDA, the classic classifier after CSP [lecture p.60], trained on our features before building HWNet. Same 5 folds as the real pipeline: fit on the training fold, score on the validation fold of session T, mean of 9 subjects (band = ±1 SE). Session E not used: not test results, not comparable with MixNet.

What the model sees: 36 features × 15 windows



- Each cell: right minus left class mean, in training SD units
- Strongest in 8-12 Hz (Mu) and 20-24 Hz (beta), from about 0.5 to 2.5 s, the bands where ERD/ERS shows up [lecture p.51]
- Rows alternate in sign because CSP takes filters from both ends: one end has more power for left hand, the other for right [lecture p.57]

Takeaway

Each trial becomes *which band*, *which spatial filter* and *when*, in 540 numbers.

All 144 session T trials per subject, averaged over 9 subjects. Session E not used.

Backup: settings and data notes

All settings live in `experiments/prep_HW.py`

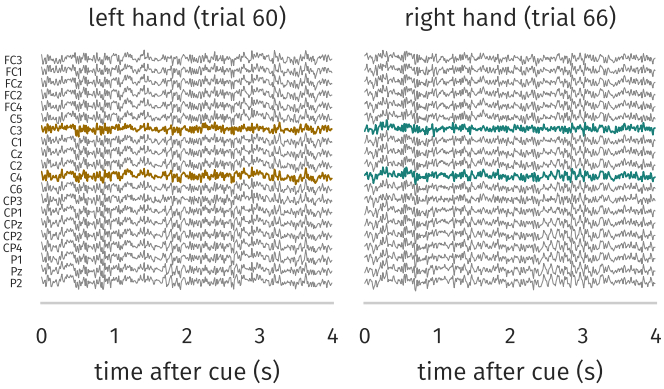
Setting	Ours	Ablation
`win_len`, `win_step`	0.5 s, 0.25 s	4.0 s, 4.0 s (classic)
`n_components`	4	2 (MixNet size)
`norm_logvar`	True	False
`bands`	9 bands, 4-40 Hz	
`order`	5	

Quick LDA check (validation, session T): ours 84.7%, classic 82.0%, `norm_logvar=False` 85.0%, `n_components=2` not run yet.

Data notes

- 50 Hz line noise: removed. After resampling to 100 Hz it sits at the top edge, where every band filter blocks it
- Drift and eye movement below 4 Hz: removed by the lowest band edge
- Not removed: blinks in 4-8 Hz, muscle noise in beta, and trials marked as artifacts (2 to 36 of 144 per subject)

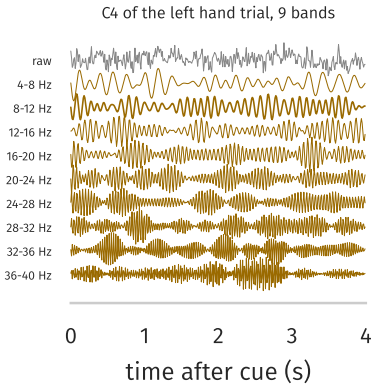
S3 walkthrough (1/5): raw EEG, 20 × 400

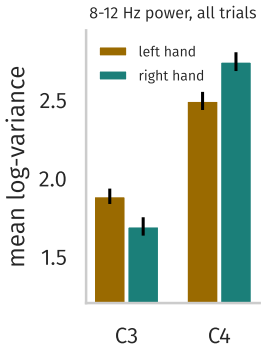


- One left hand and one right hand trial: 20 channels, 4 s at 100 Hz
- C3 and C4 (over the hand areas) in colour
- By eye the two classes look the same: the difference is hidden in band power

Subject 3, session T only (144 trials, 72 per class). Example trials = the trial closest to its class mean. CSP and scaler fit on all session T trials of S3 for this picture; no accuracy is computed here.

S3 walkthrough (2/5): filter bank, $9 \times 20 \times 400$

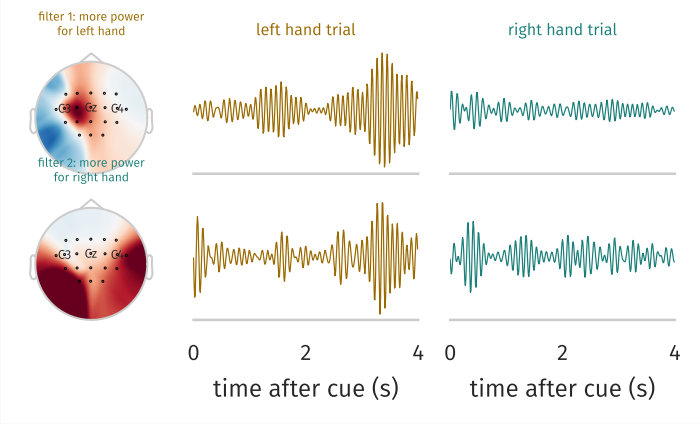




- Left: C4 of the left hand trial split into 9 bands (each trace scaled on its own)
- Right: 8-12 Hz power is lower at C4 for left hand and lower at C3 for right hand
- That is contralateral ERD [lecture p.51]

Subject 3, session T only (144 trials, 72 per class). Example trials = the trial closest to its class mean. CSP and scaler fit on all session T trials of S3 for this picture; no accuracy is computed here.

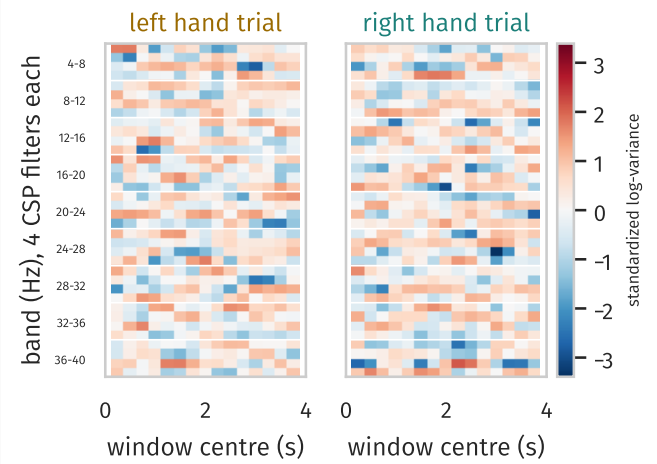
S3 walkthrough (3/5): CSP per band, 36×400 (8-12 Hz shown)



- Maps: spatial pattern of each filter, i.e. where it looks [lecture p.58]
- Filter 1 gives more power for left hand, filter 2 for right hand
- Filter 1 shows the gap clearly in these two trials; filter 2 less so, single trials are noisy
- 20 channels become 4 signals per band; time is untouched

Subject 3, session T only (144 trials, 72 per class). Example trials = the trial closest to its class mean. CSP and scaler fit on all session T trials of S3 for this picture; no accuracy is computed here.

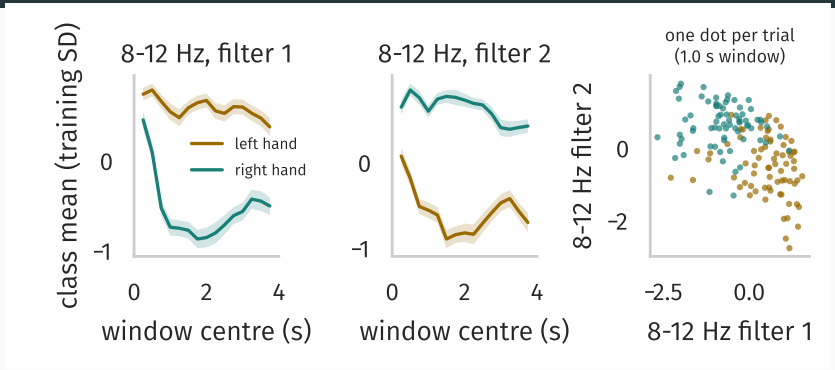
S3 walkthrough (4/5): windowed log-variance, 36 × 15



- The same two trials as the model sees them: 36 rows (9 bands × 4 filters) × 15 windows, standardized
- One trial alone is noisy; the class difference shows up only across many trials (next slide)

Subject 3, session T only (144 trials, 72 per class). Example trials = the trial closest to its class mean. CSP and scaler fit on all session T trials of S3 for this picture; no accuracy is computed here.

S3 walkthrough (5/5): do the features separate the classes?



- Class means of the two 8-12 Hz filters: the gap is small right after the cue and grows within about 1 s, the timing our windows keep
- Right: every trial at the 1.0 s window, using only 2 of the 540 numbers, already mostly separable
- S3 is one of the easier subjects (quick LDA check: 95.8%), so other subjects look less clean

Subject 3, session T only (144 trials, 72 per class). Example trials = the trial closest to its class mean. CSP and scaler fit on all session T trials of S3 for this picture; no accuracy is computed here.